

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12418233
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Zou; Qiong et al.

Boost inductor recirculation protection based on switch node detection method

Abstract

Systems and methods are provided to supply a voltage (V_{cc}) to a primary IC in a switch-mode-power-supply following startup. Briefly, an input voltage ($V_{sub.IN}$) coupled through a boost-inductor to a switch-pin of the IC is sensed by isolating a DC component ($V_{sub.DC_BSW}$) of the voltage and checking if $V_{sub.IN}$ is greater than a minimum $V_{sub.IN}$ ($V_{sub.IN_MIN}$) by comparing $V_{sub.DC_BSW}$ to a starting reference voltage ($V_{sub.REF_START}$) in the IC. If $V_{sub.DC_BSW}$ is greater than $V_{sub.REF_START}$, a boost-switch in the IC through which the switch-pin is coupled to ground is cycled on and off to alternately store and discharge energy in the boost-inductor, boosting V_{cc} . Thereafter, $V_{sub.IN}$ is checked against a maximum input voltage by comparing $V_{sub.DC_BSW}$ to a reference voltage ($V_{sub.REF}$), greater than $V_{sub.REF_START}$, and if $V_{sub.DC_BSW}$ is greater than $V_{sub.REF}$, boost is turned off by turning off and ceasing to cycle the boost-switch, protecting the boost-inductor from current runaway.

Inventors:	Zou; Qiong (Singapore, SG), Tan; Soon Hwei (Singapore, SG)
Applicant:	Cypress Semiconductor Corporation (San Jose, CA)
Family ID:	1000008820008
Assignee:	Cypress Semiconductor Corporation (San Jose, CA)
Appl. No.:	18/322419
Filed:	May 23, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240063719 A1	Feb. 22, 2024

Related U.S. Application Data

Publication Classification

Int. Cl.: **H02M1/36** (20070101); **H02M1/00** (20070101); **H02M1/32** (20070101); **H02M3/156** (20060101); **H02M3/335** (20060101)

U.S. Cl.:

CPC **H02M1/36** (20130101); **H02M1/0006** (20210501); **H02M3/33507** (20130101); **H02M1/32** (20130101); **H02M3/156** (20130101)

Field of Classification Search

CPC: H02M (1/0006); H02M (1/36); H02M (3/155-1588); H02M (3/335-33592)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8289731	12/2011	Spiridon	363/21.16	H02M 3/335
2020/0127575	12/2019	Liu	N/A	G06F 1/266

OTHER PUBLICATIONS

“UCC28782 High-Density Active-Clamp Flyback Controller with EMI Dithering, X-Cap Discharge, and Bias Power Management”, Texas Instruments Incorporated, May 2020—Revised May 2021, 98 pages. cited by applicant

Primary Examiner: Finch, III; Fred E

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is claims the benefit of priority under 35 U.S.C. 119(e) to U.S. Provisional Patent Application Ser. No. 63/399,359, filed Aug. 19, 2022, which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

(1) This disclosure relates generally to switch-mode power supplies, and more particularly to universal serial bus power delivery (USB-PD) power supply and methods of operating the same to provide stable power to a primary integrated circuit in a controller thereof.

BACKGROUND

(2) Universal Serial Bus Power Delivery (USB-PD) power supplies are one type of a switch-mode power supply (SMPS) designed for a wide range of output power applications. For example, USB-PD applications satisfying Standard Power Range (SPR) specifications can provide voltages from 3.3V to 20V with output powers up to 100 W, and applications satisfying Extended Power Range (EPR) standards can provide voltages up to 48V with output powers up to 240 W.

(3) FIG. 1A shows a schematic of one type of SMPS suitable for use as a USB-PD adapter known

as a fly-back converter **100**. Briefly, the fly-back converter **100** includes a transformer **102**, a first rectifying element or power switch (PS **104**), on a primary side of the transformer, and a second rectifying element, such as a synchronous rectifier (SR) field effect transistor (SR_FET **106**) on a secondary side of the transformer coupled between the transformer **102** and a secondary output capacitor **108** across which an output voltage is generated. The fly-back converter **100** further includes a primary side controller **110** coupled to a gate of the PS **104** and a secondary side or SR controller **112** coupled to a gate of the SR_FET **106**.

(4) When the PS **104** is closed the primary side of the transformer is connected to a voltage source coupled to the input and resulting in a primary current and magnetic flux, storing energy in the transformer **102**. When the PS **104** is opened by the primary side controller **110** and the SR_FET **106** is closed by the SR controller **112**, the energy stored in the transformer generates a secondary current, recharging the capacitor **108** supplying an output voltage to the output. Generally, the converter further includes a feedback circuit or signal path **114**, such as a pulse width transformer, coupled between the primary side controller **110** and the secondary side controller **112** to synchronize operation of the PS **104** and SR_FET **106** and to turn on or off the PS based on the voltage or power required on the output capacitor **108**.

(5) The fly-back converter **100** can be used in both AC/DC, and DC/DC conversion. Where an AC input voltage is used as the voltage source, the fly-back converter **100** further includes a rectifying circuit, such as a bridge rectifier **116**, coupled between the voltage source and the input.

(6) In certain embodiments, such as that shown, the primary side controller **110** is implemented partly or wholly as an integrated circuit or primary IC **118**. The primary IC **118** includes a gate driver **120** coupled to the gate of the PS **104** and is operable to turn the PS on and off. Typically, the primary side controller **110** further includes a startup circuit **122** to provide a startup voltage to power the primary IC **118** directly from the input during a softstart of the fly-back converter, and a primary supply circuit **124** to provide a supply voltage (V_{cc}) to the primary IC from an input voltage (V_{sub.IN}) which is proportional to a voltage generated on the secondary side of the transformer upon completion of the softstart.

(7) The startup circuit **122** includes a high voltage (HV) startup circuit or module **126** in the primary IC **118** coupled to the input of the fly-back converter **100** to provide power at a proper voltage to the primary IC. Where, as in the embodiment shown, the voltage source supplied to the input of the fly-back converter **100** is an AC voltage source, the startup circuit **122** further includes a number of diodes or rectifying elements **128** coupled between the input and the HV module **126** to provide a high DC voltage to the HV module.

(8) The primary supply circuit **124** generally includes a boost-inductor **130** coupled between an auxiliary winding **132** of the transformer **102** and through a boost-switch pin (BSW-pin) to a boost controller **134** in the primary IC **118**. A first end or terminal of the boost-inductor **130** is coupled to the auxiliary winding **132** through a low pass filter **136** formed by a capacitor and a diode, and a second end or terminal is coupled to the BSW-pin. The second terminal of the boost-inductor **130** is further coupled to a voltage supply-pin (V_{cc}-pin) in the primary IC **118** through a diode **138** connected between the BSW-pin and V_{cc}-pin, and to ground through a capacitor **140** connected between the V_{cc}-pin and ground. The boost controller **134** includes a boost-switch **142** through which the BSW-pin can be coupled to a boost ground pin (BGND-pin) to lower a voltage (V_{sub.BSW}) on the BSW-pin and a store energy in the boost-inductor **130**. The boost controller **134** is operable to cycle the boost-switch **142** on and off to store energy in the boost-inductor **130** when the boost-switch is on and to boost the voltage supplied to the V_{cc}-pin when the switch is off.

(9) Referring to FIG. 1B, one problem with previous generations of fly-back converters, such as that shown in FIG. 1A, is that as an input voltage (V_{sub.IN} **144**) from the auxiliary winding **132** approaches and exceeds a boost converter regulation voltage **146** provided by the boost-controller **134** and boost-inductor **130**, the time for which the boost-switch **142** is off (illustrated by line **148**) is increased, which causes inductor current (illustrated by line **150**) to recirculate and not fully

discharge leading to current runaway and possible damage to the boost-inductor **130** and/or primary IC **118** when an inductor saturation current is exceeded. In addition, if $V_{sub.IN}$ is insufficiently high, efficiency of boost would become very low and thus enabling boost regulator would in-turn wasting system energy.

(10) Past approaches to solving the above problem have required additional connections to the auxiliary winding **132** and internal sensing circuitry to sense $V_{sub.IN}$, resulting in the need for additional-pins on the primary IC **118** increasing a die-size of the primary IC, and increasing quiescent currents as a result of the additional sensing circuitry.

(11) Accordingly, there is a need for a primary supply circuit and methods for providing power to a primary IC in a SMPS or USB-PD that is capable of sensing input voltages ($V_{sub.IN}$) and protecting the boost-inductor and primary IC from damage due to current runaway as well as enabling boost only when input voltage ($V_{sub.IN}$) is sufficiently high for optimum boost efficiency. It is further desirable that the circuit and method are able to do so without increasing the quiescent current, and size or complexity of the primary side controller.

SUMMARY

(12) A primary supply circuit including a boost converter and methods for operating the same to provide power to a primary integrated circuit (IC) in a switch mode power supply (SMPS), such as a Universal Serial Bus power delivery (USB-PD) charger or adapter are disclosed. Briefly, the method begins after softstart of the SMPS with sensing an input voltage ($V_{sub.IN}$) coupled through a boost-inductor to a switch-pin of the IC and through a rectifier or diode from the BSW-pin to a voltage supply-pin (V_{cc} -pin). The V_{cc} -pin is further coupled to ground through a capacitor to generate a supply voltage (V_{cc}) on the V_{cc} -pin. $V_{sub.IN}$ is checked to determine if $V_{sub.IN}$ greater than or equal to a minimum input voltage ($V_{sub.IN_MIN}$). Generally, $V_{sub.IN}$ is sensed by isolating a DC component ($V_{sub.DC_BSW}$) of a switching voltage ($V_{sub.BSW}$) on the BSW-pin, and comparing $V_{sub.DC_BSW}$ to a starting reference voltage ($V_{sub.REF_START}$) in the primary IC. If $V_{sub.DC_BSW}$ is greater than or equal to $V_{sub.REF_START}$, a boost-converter is operated to turn on boost or boost-mode by cycling a boost-switch in the primary IC through which the BSW-pin is coupled to ground on and off to alternately store and discharge energy in the boost-inductor, boosting V_{cc} . If $V_{sub.IN}$ is not greater than or equal to $V_{sub.IN_MIN}$, the boost-converter is operated to confirm that boost is turned off by operating the boost-converter to turn off the boost-switch, while powering the primary IC with a startup circuit, and repeating checking if $V_{sub.IN}$ is greater than or equal to $V_{sub.IN_MIN}$.

(13) After turning on boost $V_{sub.IN}$ is checked to determine if it has risen to greater than or equal to a maximum input voltage ($V_{sub.IN_MAX}$) by comparing $V_{sub.DC_BSW}$ to another reference voltage ($V_{sub.REF}$) in the primary IC, where $V_{sub.REF}$ is greater than $V_{sub.REF_START}$, and if $V_{sub.DC_BSW}$ is greater than or equal to $V_{sub.REF}$, the boost-converter is operated to turn off boost by turning off and ceasing to cycle the boost-switch on and off, protecting the boost-inductor from damage due to current runaway. After the boost has been turned off $V_{sub.IN}$ is checked to determine if $V_{sub.IN}$ has decreased to less than or equal to a $V_{sub.IN}$ under voltage level ($V_{sub.IN_UV}$), and if $V_{sub.IN}$ has decreased to less than or equal $V_{sub.IN_UV}$, $V_{sub.IN}$ is again checked to determine if it is greater than or equal $V_{sub.IN_MIN}$, and if $V_{sub.IN}$ is greater than or equal $V_{sub.IN_MIN}$ boost is turned back on by resuming cycling the boost-switch on and off.

(14) Further features and advantages of embodiments of the invention, as well as the structure and operation of various embodiments of the invention, are described in detail below with reference to the accompanying drawings. It is noted that the invention is not limited to the specific embodiments described herein. Such embodiments are presented herein for illustrative purposes only. Additional embodiments will be apparent to a person skilled in the relevant art(s) based on the teachings contained herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1A is a simplified schematic block diagram illustrating a switch mode power supply (SMPS) including a primary side controller having a primary integrated circuit (IC) powered by a conventional primary supply circuit following an initial startup;
- (2) FIG. 1B are graphs illustrating input voltage to and current through a boost-inductor in the conventional boost converter of FIG. 1A;
- (3) FIG. 2A is a schematic block diagram illustrating a SMPS including a primary IC powered following an initial startup by a boost-converter according to the present disclosure;
- (4) FIG. 2B is a schematic block diagram illustrating the boost-converter of FIG. 2A;
- (5) FIG. 3 are graphs illustrating input voltage ($V_{sub.IN}$) and a supply voltage (V_{cc}) supplied to a primary IC by boost-converter of FIGS. 2A and 2B following an initial startup; and
- (6) FIG. 4 is a flowchart illustrating a method for supplying stable power to a primary IC in a SMPS following an initial startup.

DETAILED DESCRIPTION

- (7) A system and methods are provided for supplying stable power to a primary integrated circuit (IC) in a switch-mode power supply (SMPS), such as a Universal Serial Bus power delivery (USB-PD) application following an initial startup. Briefly, the system and method are based on preventing damage to a boost-inductor in a boost-converter by sensing an over voltage on a boost-switch node or pin (BSW-pin) without the need for any additional-pins.
- (8) In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the present invention. It will be evident, however, to one skilled in the art that the present invention may be practiced without these specific details. In other instances, well-known structures, and techniques are not shown in detail or are shown in block diagram form in order to avoid unnecessarily obscuring an understanding of this description.
- (9) Reference in the description to “one embodiment” or “an embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the invention. The appearances of the phrase “in one embodiment” in various places in the specification do not necessarily all refer to the same embodiment. The term to couple as used herein may include both to directly electrically connect two or more components or elements and to indirectly connect through one or more intervening components.
- (10) An embodiment of a system for supplying power to a primary IC in a SMPS following an initial startup by a primary supply circuit including or consisting of a boost converter will now be described with reference to FIGS. 2A and 2B. FIG. 2A is a schematic block diagram illustrating one type of a SMPS particularly useful in a USB-PD adapter or charger, and including a fly-back converter with a primary IC powered following an initial startup by a primary supply circuit including a boost converter.
- (11) Referring to FIG. 2A, the fly-back converter **200**, and includes a transformer **202**, a first rectifying element or power switch (PS **204**), such as a metal-oxide-silicon field effect transistor (MOSFET), on a primary side of the transformer, a second rectifying element, such as a synchronous rectifier (SR) field effect transistor (SR_FET **206**), coupled between a secondary side of the transformer and as secondary output capacitor **208**, a primary side controller **210** coupled to a gate of the PS **204**, and a secondary side controller **212** coupled to a gate of the SR_FET **206**. The input voltage to the converter can include either an AC input voltage or a sinusoidal DC input voltage coupled to the primary side of the transformer **202** through a rectifying circuit, such as a bridge rectifier **214**, and across a primary input capacitor **215** as in the embodiment shown. A pulse transformer **216** coupled between the primary side controller **210** and the secondary side controller **212** provides a pulse width modulated (PWM) signal based on a DC voltage or power required on

the output capacitor **208** to turn on or off the PS **204**.

(12) In certain embodiments, such as that shown, the primary side controller **210** is implemented partly or wholly as an integrated circuit or primary IC **218**. The primary IC **218** includes a gate driver **220** coupled to the gate of the PS **204** and is operable to turn the PS on and off in response to the PWM signal. In these embodiments the fly-back converter **200** typically further includes a startup circuit **222** to provide a startup voltage and power primary IC **218** during a softstart of the fly-back converter, and a primary supply circuit or boost-converter **224** to provide a supply voltage (V_{cc}) to the primary IC **218** following an initial softstart of the fly-back converter using the startup circuit **222** after a DC voltage has built-up on the secondary output capacitor **208**.

(13) The startup circuit **222** is coupled to input terminals of the fly-back converter **200**, a number of diodes or rectifying elements **228** coupled to the primary input, and a high voltage (HV) startup circuit or module **230** in the primary IC **218**.

(14) FIG. 2B is a more detailed diagram of the boost-converter **224** of FIG. 2A. Referring to FIG. 2B the boost-converter **224** includes a boost-inductor **232** coupled between a voltage input and a boost-switch node or switch-pin (BSW-pin **234**) of the primary IC **218**. The boost-inductor **232** is operable to receive an input voltage ($V_{sub.IN}$) from the voltage input and couple a switching voltage ($V_{sub.BSW}$) to the BSW-pin **234**. In the embodiment shown, the voltage input is provided by an auxiliary winding **236** of the transformer **202** coupled to the boost-inductor **232** through a first low pass filter **238** formed by a diode ($D_{sub.filter}$ **240**) and a capacitor ($C_{sub.filter}$ **241**). An external rectifier diode or rectifier **242** is coupled between the BSW-pin **234** and a voltage supply-pin (V_{cc} -pin **244**), and the V_{cc} -pin further coupled to ground through an external capacitor **246** to provide a supply voltage (V_{cc}) to power the primary IC **218** after a DC voltage has built-up on the secondary output capacitor **208**.

(15) The boost-converter **224** further includes a boost-circuit **248** integrally formed in the primary IC **218**. Generally, the boost-circuit **248** includes a comparator **250** with a non-inverting (+) input coupled to one or more internal reference voltages ($V_{sub.REF}$ and $V_{sub.REF_START}$), an inverting (−) input coupled to the BSW-pin **234** through a voltage ladder or divider **252** formed by resistors R_1 and R_2 and a second low pass filter **254** formed by a resistor $R_{sub.filter}$ and a capacitor $C_{sub.filter}$ to receive a divided and filtered DC component ($V_{sub.DC_BSW}$) of $V_{sub.BSW}$, and a boost-switch **256**. As explained further below, the $V_{sub.REF_START}$ is a starting reference voltage used immediately following softstart of the fly-back converter **200** to determine $V_{sub.DC_BSW} \geq V_{sub.REF_START}$, and therefore $V_{sub.IN}$ is greater than or equal to a minimum input voltage ($V_{sub.IN_MIN}$) required to begin cycling the boost-switch **256**, while the $V_{sub.REF}$ reference is used during boost-mode or boost-switch cycling is enabled or commenced to provide a boosted V_{cc} following softstart.

(16) The boost-switch **256** is a transistor, such as an N-channel low side field effect transistor (NMOS LSFET), and has a gate coupled to an output of the comparator **250** through a gate driver or controller **258** by which is operable to turn the boost-switch alternately on and off in boost-mode, and to turn off and cease cycling the boost-switch to turn off boost-mode. The boost-switch **256** is operable when turned on to couple the BSW-pin **234** to ground through a boost-switch ground-pin (BGND-pin **264**), lowering the switching voltage ($V_{sub.BSW}$) on the BSW-pin and storing energy in the boost-inductor **232**. When the boost-switch **256** is subsequently turned off, the boost-inductor **232** discharges boosting the switching voltage ($V_{sub.BSW}$) on the BSW-pin **234** and supplied through rectifier **242** to the V_{cc} -pin **244**.

(17) Generally, the reference voltage, $V_{sub.REF}$ is an o voltage selected based on an over-voltage protection level ($V_{sub.OVP}$), and the comparator **250** is operable to output a signal to turn off the cycling of the boost-switch **256** when the divided and filtered DC component voltage of the voltage on the BSW-pin **234**, is greater than or equal to the reference voltage, that is when $V_{sub.DC_BSW} \geq V_{sub.REF}$, to protect the boost-inductor **232** from damage due to current runaway. The comparator **250** is further operable to output a signal to turn the cycling of the boost-

switch **256** back on when V.sub.IN is dropped below a under-voltage level (V.sub.IN_UV), where V.sub.IN_UV is selected to be higher than a minimum supply voltage (Vcc.sub.MIN) for normal operation of the primary IC. In one embodiment, the boost-circuit **248** detects that V.sub.IN has dropped below V.sub.IN_UV by detecting that V.sub.DC_BSW is less than V.sub.REF by greater than or equal to an offset voltage (V.sub.OFF).

(18) In another embodiment, the comparator **250** is a Schmitt trigger comparator with a positive feedback loop from the output of the comparator to the non-inverting input, and the offset voltage is based on a hysteresis of the comparator.

(19) In addition, the controller **258** further includes a maximum time on circuit (Max T on **260**) to turn off the boost-switch **256** after it has been on for more than a maximum time on, and a maximum time off circuit (Max T off **262**) to turn on the boost-switch **256** after it has been off for more than a maximum time off.

(20) FIG. **3** includes graphs illustrating input voltage (V.sub.IN) and a supply voltage (Vcc) supplied to a primary IC by boost-converter of FIGS. **2A** and **2B** following an initial startup, and a corresponding timing diagram showing times for which the cycling of the boost-switch **256** is turned on and off. Referring to FIG. **3**, between time t0 and time t1 V.sub.IN **302** is increasing, Vcc **304** is equal to a start voltage (Vcc.sub.START) supplied by the HV startup module **230**, and the boost-switch cycling **306** is off to while waiting for V.sub.IN to rise to greater than or equal to a minimum voltage level (V.sub.IN_MIN) required to enable operation of the boost-converter **224**.

(21) At time t1 V.sub.IN has risen to V.sub.IN_MIN (V.sub.IN ≥ V.sub.IN_MIN), boost-switch cycling **306** begins and Vcc **304** begins to rise. Generally, determining V.sub.IN is greater than or equal to V.sub.IN_MIN is accomplished by comparing V.sub.DC_BSW to V.sub.REF_START. A portion **308** of the line illustrating boost-switch cycling **306** indicates times during which the boost-switch cycling is turned on, with the boost-switch **256** being alternately turned on and off to boost Vcc.

(22) From time t1 to t3 boost-switch cycling **306** is turned on with Vcc cycling beginning at time t2 as a result of the normal operation of the boost-converter **224**, and while V.sub.IN continues to rise as the output of the fly-back converter **200** rises.

(23) At time t3 V.sub.IN has risen to a maximum voltage level (V.sub.IN_MAX), which is selected to be equal to an over voltage protection (V.sub.OVP) to protect the boost-inductor **232** from damage due to current runaway. The boost-circuit **248** detects V.sub.IN ≥ V.sub.IN_MAX by sensing V.sub.DC_BSW is greater than or equal to V.sub.REF, and turns off the boost-switch cycling **306** by turning off and ceasing to cycle the boost-switch **256**. A portion **310** of the line illustrating boost-switch cycling **306** indicates times during which the boost-switch cycling is turned off, with the boost-switch **256** being turned off and ceasing to be cycled.

(24) From time t3 to time t4 Vcc **304** is maintained by V.sub.IN **302** through rectifier **242**. When during operation of the fly-back converter **200** V.sub.IN **302** and therefore Vcc **304** begin to drop due to a drop either in an input voltage to or output voltage from the fly-back converter, the boost-circuit **248** detects V.sub.IN is less than or equal to V.sub.IN_UV by sensing V.sub.DC_BSW is less than V.sub.REF by more than the offset voltage, and at time t4 the boost-circuit **248** turns the boost-switch cycling **306** back on by resuming or resuming cycling the boost-switch **256** on and off.

(25) FIG. **4** is a flowchart illustrating a method for operating a primary supply circuit of a SMPS, such as the fly-back converter **200** shown in FIG. **2A** and including the boost-converter **224** shown in FIG. **2B**, to provide a stable supply voltage (Vcc) or power to the primary IC **218** following an initial startup powered by the HV startup circuit **222**.

(26) Referring to FIG. **4**, the method begins with performing a softstart to power the primary IC directly from the input to the SMPS (softstart, step **400**). This is generally accomplished using the startup circuit **222** including rectifying elements **228** coupled across input terminals to the primary, and the HV startup module **230** in the primary IC **218**.

(27) Next, a check is preformed to determine if an input voltage ($V_{sub.IN}$) has been coupled to the boost-inductor **232** of the boost-converter **224** and is greater than a minimum input voltage ($V_{sub.IN_MIN}$, step **402**). As noted above, checking for $V_{sub.IN}$ can be accomplished by dividing and filtering the voltage ($V_{sub.BSW}$) on the BSW-pin **234** using the internal voltage divider **252** and low pass filter **254** in the primary IC **218** to isolate a DC component ($V_{sub.DC_BSW}$) of $V_{sub.BSW}$ and determining if $V_{sub.DC_BSW}$ is greater than or equal to a reference start voltage ($V_{sub.REF_START}$). If $V_{sub.IN}$ is not present ($V_{sub.IN} < V_{sub.IN_MIN}$), boost-switch cycling is turned off or maintained off (Turn OFF Boost, step **404**), and the softstart continues with the startup circuit **222** providing power to the primary IC **218** directly from the input to the flyback converter **200**, while repeating the check for $V_{sub.IN}$. If $V_{sub.IN}$ is present, the boost or the boost-circuit **248** is turned on by operating the control **258** cycling the boost-switch **256** on and off to alternately store and discharge energy in the boost-inductor to boost V_{cc} (Turn ON Boost, step **406**). It is noted that a switching frequency at which the boost-switch **256** is turned on and off is generally based on $V_{sub.IN}$, and that the switching frequency increases at lower $V_{sub.IN}$ and decreases with higher $V_{sub.IN}$.

(28) Next, while continuing to operate the boost-converter **224** in boost-mode by cycling the boost-switch **256** on and off, a check is preformed to determine if $V_{sub.IN}$ is greater than or equal to a maximum voltage level ($V_{sub.IN_MAX}$, step **408**). This can be accomplished, for example, by sensing and comparing $V_{sub.DC_BSW}$ to $V_{sub.REF}$. Because $V_{sub.REF}$ is selected based on an input voltage over-voltage protection level ($V_{sub.OVP}$), which is selected to protect the boost-inductor from damage due to current runaway, if $V_{sub.DC_BSW}$ is greater than $V_{sub.REF}$, the current through the boost-inductor **232** could damage the boost-inductor **232**. If $V_{sub.DC_BSW}$ is not greater than $V_{sub.REF}$, then $V_{sub.IN}$ is not greater than or equal to $V_{sub.IN_MAX}$, and the boost-switch cycling is maintained on with the boost-switch **256** continuing to be cycled on and off, while repeating steps **402** to **406**. If $V_{sub.DC_BSW}$ is greater than or equal to $V_{sub.REF}$ the boost-switch cycling is turned off by turning off and ceasing to cycle the boost-switch **256** (Turn OFF Boost, step **410**).

(29) Next, a check is preformed to determine if $V_{sub.IN}$ has dropped to or below a under voltage level ($V_{sub.IN_UV}$) ($V_{sub.IN} < V_{sub.IN_UV}$, step **412**). As noted above, this can be accomplished by sensing that $V_{sub.DC_BSW}$ is more than an offset voltage ($V_{sub.OFF}$) below $V_{sub.REF}$, or by a hysteresis of the comparator based on the $V_{sub.IN_UV}$ and offset voltage. If $V_{sub.IN}$ has dropped to or below $V_{sub.IN_UV}$, the steps of the above method, i.e., steps **402** to **412**, are repeated beginning with determining if $V_{sub.IN}$ is present to turn the boost-circuit **248** back on by resuming cycling of the boost-switch **256**.

(30) If $V_{sub.IN}$ has not dropped below $V_{sub.IN_UV}$ the boost-circuit **248** is maintained off with the boost-switch **256** off, while repeating the check to determine if $V_{sub.IN}$ has dropped to or below $V_{sub.IN_UV}$.

(31) Thus, systems for supplying stable power to a primary IC in a SMPS following an initial startup, and methods of operating the same to protect a boost-inductor and the primary IC from damage due to current runaway, without increasing the size, complexity and/or cost of a primary side controller have been disclosed. Embodiments of the present invention have been described above with the aid of functional and schematic block diagrams illustrating the implementation of specified functions and relationships thereof. The boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the description. Alternate boundaries can be defined so long as the specified functions and relationships thereof are appropriately performed.

(32) The foregoing description of the specific embodiments will so fully reveal the general nature of the invention that others can, by applying knowledge within the skill of the art, readily modify and/or adapt for various applications such specific embodiments, without undue experimentation, without departing from the general concept of the present invention.

(33) It is to be understood that the Detailed Description section, and not the Summary and Abstract sections, is intended to be used to interpret the claims. The Summary and Abstract sections may set forth one or more but not all exemplary embodiments of the present invention as contemplated by the inventor(s), and thus, are not intended to limit the present invention and the appended claims in any way.

(34) The breadth and scope of the present invention should not be limited by any of the above-described exemplary embodiments, but should be defined only in accordance with the following claims and their equivalents.

Claims

1. A method of operating a switch mode power supply (SMPS), the method comprising: following an initial startup of the SMPS, sensing an input voltage ($V_{sub.IN}$) coupled through a boost-inductor to a switch pin (BSW-pin) of a primary integrated circuit (IC) in the SMPS by isolating a DC component ($V_{sub.DC_BSW}$) of a switching voltage ($V_{sub.BSW}$) coupled to the BSW-pin; checking if $V_{sub.IN}$ is greater than or equal to a minimum input voltage ($V_{sub.IN_MIN}$) by comparing $V_{sub.DC_BSW}$ to a starting reference voltage ($V_{sub.REF_START}$) in the primary IC; and if $V_{sub.DC_BSW}$ is greater than or equal to $V_{sub.REF_START}$: operating a boost-converter to turn on boost-mode by cycling a boost-switch in the primary IC through which the BSW-pin is coupled to ground on and off to alternately store and discharge energy in the boost-inductor; and boosting a supply voltage (V_{cc}) to a V_{cc} -pin of the primary IC coupled to the BSW-pin and boost-inductor through a rectifier and to ground through a capacitor on which V_{cc} is generated.
2. The method of claim 1 wherein if $V_{sub.IN}$ is not greater than or equal to $V_{sub.IN_MIN}$ the method comprises confirming the boost-mode is turned off by operating the boost-converter to turn off the boost-switch, while powering the primary IC with a startup circuit, and repeating checking if $V_{sub.IN}$ is greater than or equal to $V_{sub.IN_MIN}$.
3. The method of claim 1 further comprising after turning on boost-mode checking if $V_{sub.IN}$ is greater than or equal to a maximum input voltage ($V_{sub.IN_MAX}$) by comparing $V_{sub.DC_BSW}$ to a reference voltage ($V_{sub.REF}$) in the primary IC, where $V_{sub.REF}$ is greater than $V_{sub.REF_START}$, and if $V_{sub.DC_BSW}$ is greater than or equal to $V_{sub.REF}$, the method further comprises operating the boost-converter to turn off boost-mode by turning off and ceasing to cycle the boost-switch on and off.
4. The method of claim 3 further comprising after turning off boost-mode, checking if $V_{sub.IN}$ has decreased to less than or equal to a $V_{sub.IN}$ under voltage level ($V_{sub.IN_UV}$), and if $V_{sub.IN}$ has decreased to less than or equal $V_{sub.IN_UV}$, the method further comprises checking if $V_{sub.IN}$ is greater than or equal $V_{sub.IN_MIN}$, and if $V_{sub.IN}$ is greater than or equal $V_{sub.IN_MIN}$ turning boost-mode back on by resuming cycling the boost-switch on and off.
5. The method of claim 4 wherein checking to determine if $V_{sub.IN}$ has decreased to less than or equal to $V_{sub.IN_UV}$ comprises comparing $V_{sub.DC_BSW}$ to $V_{sub.REF}$ and determining if $V_{sub.DC_BSW}$ is more than an offset voltage ($V_{sub.OFF}$) below $V_{sub.REF}$.
6. The method of claim 5 wherein if $V_{sub.IN}$ has not decreased to less than or equal $V_{sub.IN_UV}$ the method further comprises repeating checking to determine if $V_{sub.IN}$ has decreased to less than or equal to $V_{sub.IN_UV}$.
7. The method of claim 5 wherein comparing $V_{sub.DC_BSW}$ to $V_{sub.REF}$ comprises coupling $V_{sub.DC_BSW}$ to an inverting input of a comparator in the primary IC while coupling $V_{sub.REF}$ to a non-inverting input of the comparator, and wherein the $V_{sub.OFF}$ is provided by a hysteresis of the comparator implemented by a positive feedback loop from an output of the comparator to the non-inverting input.
8. The method of claim 1 wherein comparing $V_{sub.DC_BSW}$ to $V_{sub.REF_START}$ comprises comparing $V_{sub.DC_BSW}$ to $V_{sub.REF_START}$ using a comparator in the primary IC, and

wherein isolating $V_{sub.DC_BSW}$ comprises dividing and filtering $V_{sub.BSW}$ using a voltage divider and a low pass filter in the primary IC through which the BSW-pin is coupled to an input of the comparator.

9. A boost-converter comprising: a boost-inductor coupled between a voltage input and a boost-switch pin (BSW-pin) of a primary integrated circuit (IC) in a switch mode power supply (SMPS), the boost-inductor operable to receive an input voltage ($V_{sub.IN}$) and couple a switching voltage ($V_{sub.BSW}$) to the BSW-pin; a voltage supply-pin (V_{cc} -pin) in the primary IC coupled to the BSW-pin and the boost-inductor through a diode, the V_{cc} -pin further coupled to ground through a capacitor to generate a supply voltage (V_{cc}) on the V_{cc} -pin; and a boost-circuit in the primary IC coupled to the BSW-pin, and including a controller coupled to a boost-switch through which the BSW-pin is coupled to ground, the controller operable to turn on boost-mode by cycling the boost-switch on and off to alternately store energy in and discharge the boost-inductor boosting V_{cc} , wherein the boost-circuit is further operable to sense $V_{sub.IN}$, determine if $V_{sub.IN}$ is greater than or equal to a maximum input voltage ($V_{sub.IN_MAX}$), and if $V_{sub.IN}$ is greater than or equal to $V_{sub.IN_MAX}$ to protect the boost-inductor from damage due to current runaway by turning off boost-mode by turning off and ceasing to cycle the boost-switch.

10. The boost-converter of claim 9 wherein the boost-circuit further includes a comparator having an inverting input coupled to the BSW-pin through an internal voltage divider and a low pass filter, and a first non-inverting input coupled to an internal reference voltage ($V_{sub.REF}$), and wherein the boost-circuit is operable to determine if $V_{sub.IN}$ is greater than or equal to $V_{sub.IN_MAX}$ by isolating a DC component ($V_{sub.DC_BSW}$) of a switching voltage ($V_{sub.BSW}$) on the BSW-pin by dividing and filtering $V_{sub.BSW}$ using the voltage divider and low pass filter, and to compare $V_{sub.DC_BSW}$ to $V_{sub.REF}$ using the comparator.

11. The boost-converter of claim 10 wherein the boost-circuit is further operable to sense if $V_{sub.IN}$ has decreased to less than or equal to an under voltage level ($V_{sub.IN_UV}$) by determining if $V_{sub.DC_BSW}$ is greater than or equal to an offset voltage ($V_{sub.OFF}$) below $V_{sub.REF}$, and if $V_{sub.IN}$ is less than or equal to $V_{sub.IN_UV}$ to sense if $V_{sub.IN}$ is greater than or equal to a minimum input voltage ($V_{sub.IN_MIN}$), and if $V_{sub.IN}$ is greater than or equal to $V_{sub.IN_MIN}$ to turn the boost-mode back on by resuming cycling the boost-switch on and off.

12. The boost-converter of claim 11 wherein the $V_{sub.OFF}$ is based on a hysteresis of the comparator implemented by a positive feedback loop from an output of the comparator to the non-inverting input to provide the $V_{sub.OFF}$.

13. The boost-converter of claim 10 wherein the comparator further comprises a second non-inverting input coupled to a starting reference voltage ($V_{sub.REF_START}$), and wherein the boost-circuit is further operable following an initial startup of the SMPS to sense if $V_{sub.IN}$ is greater than or equal to a minimum input voltage ($V_{sub.IN_MIN}$), by determining if $V_{sub.DC_BSW}$ is greater than or equal to $V_{sub.REF_START}$, and to turn on boost-mode by cycling the boost-switch on and off to alternately store and discharge energy in the boost-inductor, boosting V_{cc} .

14. The boost-converter of claim 9 wherein the SMPS is a universal serial bus power delivery (USB-PD) power supply including a transformer comprising an auxiliary winding, and wherein the boost-inductor is coupled to an output of the auxiliary winding to receive $V_{sub.IN}$ therefrom.

15. A switch mode power supply (SMPS) comprising: a transformer including a primary side coupled to receive an input voltage and having a power switch (PS) coupled between the input voltage and the primary side, and a secondary side coupled to an output to provide an output voltage thereon; a primary side controller operable to control the PS, the primary side controller including: a primary integrated circuit (IC) having a switch pin (BSW-pin) and a voltage supply-pin (V_{cc} -pin) through which a supply voltage (V_{cc}) is supplied to power the primary IC; and a boost-converter comprising: a boost-inductor coupled between an auxiliary winding in the transformer and the BSW-pin, the boost-inductor operable to receive an input voltage ($V_{sub.IN}$) and couple a switching voltage ($V_{sub.BSW}$) to the BSW-pin; a diode through which the V_{cc} -pin is coupled to

the boost-inductor and the BSW-pin; a capacitor through which the Vcc-pin is coupled to ground to generate Vcc; and a boost-circuit in the primary IC coupled to the BSW-pin, and including a controller coupled to a boost-switch through which the BSW-pin is coupled to ground, the controller operable to turn on boost-mode by cycling the boost-switch on and off to alternately store energy in and discharge the boost-inductor boosting Vcc, wherein the boost-circuit is further operable to sense V.sub.IN, determine if V.sub.IN is greater than or equal to a maximum input voltage (V.sub.IN_MAX), and if V.sub.IN is greater than or equal to V.sub.IN_MAX to protect the boost-inductor from damage due to current runaway by turning off boost-mode by turning off and ceasing to cycle the boost-switch.

16. The SMPS of claim 15 wherein the boost-circuit further includes a comparator having an inverting input coupled to the BSW-pin through an internal voltage divider and a low pass filter, and a first non-inverting input coupled to an internal reference voltage (V.sub.REF), and wherein the boost-circuit is operable to determine if V.sub.IN is greater than or equal to V.sub.IN_MAX by isolating a DC component (V.sub.DC_BSW) of a switching voltage (V.sub.BSW) on the BSW-pin by dividing and filtering V.sub.BSW using the voltage divider and low pass filter, and to compare V.sub.DC_BSW to V.sub.REF using the comparator.

17. The SMPS of claim 16 wherein the boost-circuit is further operable to sense if V.sub.IN has decreased to less than or equal to an under voltage level (V.sub.IN_UV) by determining if V.sub.DC_BSW is greater than or equal to an offset voltage (V.sub.OFF) below V.sub.REF, and if V.sub.IN is less than or equal to V.sub.IN_UV to sense if V.sub.IN is greater than or equal to a minimum input voltage (V.sub.IN_MIN), and if V.sub.IN is greater than or equal to V.sub.IN_MIN to turn the boost-mode back on by resuming cycling the boost-switch on and off.

18. The SMPS of claim 17 wherein the V.sub.OFF is based on a hysteresis of the comparator implemented by a positive feedback loop from an output of the comparator to the non-inverting input to provide the V.sub.OFF.

19. The SMPS of claim 16 wherein the comparator further comprises a second non-inverting input coupled to a starting reference voltage (V.sub.REF_START), and wherein the boost-circuit is further operable following an initial startup of the SMPS to sense if V.sub.IN is greater than or equal to a minimum input voltage (V.sub.IN_MIN), by determining if V.sub.DC_BSW is greater than or equal to V.sub.REF_START, and to turn on boost-mode by cycling the boost-switch on and off to alternately store and discharge energy in the boost-inductor, boosting Vcc.

20. The SMPS of claim 15 wherein the SMPS is a universal serial bus power delivery (USB-PD) power supply.
